Aryan Varshney

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Summary

AI/ML Engineer with hands-on experience building production Generative AI systems — real-time voice pipelines, RAG applications, and computer vision — using Python, LangChain, FAISS, and LLM APIs (LLaMA, Gemini). Strong foundations in DSA and end-to-end ML model development with Scikit-learn.

Experience

EduLinkUp — Remote

AI/ML Intern — Generative AI (RAG, LLMs, FAISS, Prompt Engineering) — *Jan 2026 – Feb 2026*

- Built a Generative AI cybersecurity tool using Gemini 2.5 Flash with **RAG** over a **FAISS vector index** of **1,800+ document pages and 10 web sources**, enabling accurate, grounded responses to cybersecurity queries.

Projects

Drishti : Swachh Bharat | | *Python, YOLOv8, Facenet512, FAISS, MediaPipe, SQLite*

- Built a real-time AI littering surveillance pipeline that detects garbage drops via **YOLOv8**, crops the violator's face using **MediaPipe**, and dispatches an automated **e-challan (500 fine)** via SMTP — all without human intervention, achieving **zero false triggers** on legitimate dustbin disposal.
- Designed a **5-stage behavioral state machine** (Untracked → Attached → Detaching → Monitoring → Confirmed) that analyses how a person interacts with garbage over time, eliminating false alarms from people simply walking near trash or disposing waste in dustbins.
- Engineered **resolution-agnostic detection thresholds** scaled to each person's on-screen height, ensuring consistent accuracy across 720p webcams and 4K CCTV cameras, with **FAISS vector search** over facial embeddings for fast identity matching.

Dr. Chakshu | | *Python, PyTorch, DenseNet-121, Grad-CAM, FastAPI, FAISS, Groq Api*

- Built a **multi-stage AI diagnostic pipeline** for chest X-rays using a fine-tuned DenseNet-121 backbone trained on **57K+ NIH CXR-14** images, achieving a mean **AUC-ROC of 0.7735** across Cardiomegaly, Pleural Effusion, and Pneumothorax with patient-aware splits to eliminate data leakage.
- Engineered an explainability layer via **Grad-CAM** exposed through a **FastAPI** endpoint that returns the highest-probability disease prediction alongside a Base64-encoded heatmap overlay, highlighting model attention regions on the denseblock4 convolutional layer.
- Designed a **RAG-augmented report generation** module using **FAISS** as a local vector store with sentence-transformers embeddings and the **Groq API** as the LLM, grounding AI-drafted structured radiology reports in retrieved medical-guideline context for clinical accuracy.

Stock Market Voice | | *Python, FastAPI, LangGraph, Groq, Twilio, Sarvam AI, Supabase, WebSocket*

- Built an **AI Voice Agent** that makes **automated concurrent outbound phone calls** to clients with personalised stock portfolio summaries; implemented a **LangGraph ReAct agent** with **4 live market data tools** (live price, metrics, history, market status) and **zero-hallucination architecture** by injecting per-client trade data from **Supabase** into the LLM system prompt at runtime.
- Developed an end-to-end **real-time voice pipeline** (Twilio Media Streams → Silero VAD → Sarvam Speech-to-Text → LLM → Sarvam Text-to-Speech → WebSocket) supporting **multiple languages** with streaming TTS playback that begins within the first audio chunk; engineered a **barge-in interruption system** detecting user speech in **300ms** that instantly cancels in-flight audio and resets the conversation state.
- Implemented robust **multi-turn conversational memory** using LangGraph **InMemorySaver** with per-call thread isolation, and designed LangGraph state pruning via **RemoveMessage** to recover gracefully from mid-tool-execution interruptions, ensuring zero agent crashes across concurrent calls.

Education

Jaypee Institute of Information Technology — Noida, India

B.Tech. in Computer Science and Engineering — *2024 – 2028*

Technical Skills

Programming Languages: Python, Java, Dart, C, C++
Frameworks: Django, FastAPI, Flask, Flutter, Spring Boot
Databases: PostgreSQL, MySQL, MongoDB
AI/ML: Machine Learning, Scikit-learn, PyTorch, HuggingFace, Embeddings, FAISS, NLP, LangChain, RAG
Tools: GitHub, Docker, REST APIs, WebSockets, n8n, Gemini API, Sarvam AI, Groq, Silero VAD

Achievements

Keploy Certified — August 2025

- **Top 168 of 18,000+ participants (top 1%)** in Keploy API Fellowship — selected through API design, testing, and documentation rounds